

Vincent Han

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EDUCATION

University of Illinois Urbana-Champaign

Champaign-Urbana, IL

Professional Masters of Computer Science

Aug. 2024 – May 2025

- **Relevant Coursework:** Deep Learning, Statistical Reinforcement Learning, Adv. Topics in Natural Language Processing, Deep Generative Models, Topics in LLM Agents, Computational Complexity

Bowdoin College

Brunswick, ME

B.A. in Computer Science, B.A. in Government and Legal Studies, English minor

Sep. 2020 – May 2024

EXPERIENCE

Full Stack Developer & CTO at OpenQuest

August 2026 – Present

Remote

- Developing a full-stack web and mobile travel platform with a custom recommendation algorithm matching users
- Built an interactive world-map visualization using React and TypeScript to dynamically render user locations
- Designed a PostgreSQL and Redis data layer for persistent storage of user profiles and graph-structured user data

Independent Machine Learning Researcher

April 2026 – Present

Remote, collaborating with a UIUC PhD Student

- Constructed a geometric manifold framework for the drifting model proposed by Deng et al. (arxiv 2602.04770)
- Developing an Onsager-geometric formulation of Drifting Models in which the neural tangent kernel defines a distribution-dependent mobility operator.
- Preprint of paper submitting on Arxiv; intending to publish on ICLR and other top ML journals

PROJECTS

Reproduction of Attention Residuals | *PyTorch*

March 2026 – April 2026

- Inspired by the paper "Attention Residuals" by Kimi Team (arxiv 2603.15031)
- Implemented Block Attention Residuals in a 12-layer, 54M-parameter causal Transformer with top-2 MoE routing
- AttnRes reduced validation perplexity from **245.50** to **198.27** relative to standard residual connections.

Language Diffusion Model Safety Steering | *PyTorch*

March 2025 – May 2025

- Applied classifier-free score steering during diffusion sampling to suppress unsafe token trajectories
- Derived a discrete analogue of the continuous score-function metric for steering token-generation trajectories
- Reduced average unsafe ratio from **0.66** to **0.60** using a novel method of steering both at initial and reverse phase

AI OpsLab: Expanded Framework and Novel Evaluation Metrics | *Hugging Face*

March 2025 – May 2025

- Utilized PPO trainer from Huggingface to train large language model to output better fault injection scenarios
- Devised novel metrics for two types of fault injection tasks: localization and root cause analysis
- Used sentence transformer from Huggingface to evaluate the semantic similarity between two outputs
- Attained a final PPO loss of **1.78** with 1000 iterations of training

Natural Language to First Order Logic Translation | *Python, Hugging Face*

October 2024 – December 2024

- Co-developed an iterative learning-to-rank method for translating natural-language entailment problems into first-order logic.
- Used DPO trainers from hugging face transformer learning model to improve the entailment preservation rate of translations — a self-devised metric of translation accuracy
- Improved EPR by **1.8–2.7%** and EPR@16 by **17.4–20.6%** across three datasets.

Bayesian Reinforcement Learning in Real Time Trading Environment | *PyTorch*

April 2024 – May 2024

- Implemented a trading agent using Bayesian Neural Network trained with variational inference, Bayes by backprop
- Compared with other machine learning or algorithmic trading agents, such as ANN networks, simple agents
- Achieved significantly higher cumulative returns (**125-130%**) than baseline ANN and heuristic agents

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, C#, HTML/CSS, SQL, TypeScript

ML Frameworks: PyTorch, Hugging Face Transformers/TRL, TensorFlow, scikit-learn, NumPy, pandas

Web: React, Node.js, MySQL, PostgreSQL, Redis, jQuery

Spoken languages: English (native), Mandarin Chinese (native), French (fluent), German (intermediate)